

Regla de la cadena para derivadas de Wirtinger

Proposición 1 (Regla de la cadena). Sean $f \in \mathcal{C}^1(\Omega_1)$ y $g \in \mathcal{C}^1(\Omega_2)$ con $f(\Omega_1) \subset \Omega_2$, entonces se satisfacen

$$(1) \quad \frac{\partial}{\partial z}(g \circ f)(z) = \frac{\partial g}{\partial z}(f(z)) \frac{\partial f}{\partial z}(z) + \frac{\partial g}{\partial \bar{z}}(f(z)) \frac{\partial \bar{f}}{\partial z}(z).$$

$$(2) \quad \frac{\partial}{\partial \bar{z}}(g \circ f)(z) = \frac{\partial g}{\partial z}(f(z)) \frac{\partial f}{\partial \bar{z}}(z) + \frac{\partial g}{\partial \bar{z}}(f(z)) \frac{\partial \bar{f}}{\partial \bar{z}}(z).$$

Demostración: Veremos únicamente la primera identidad, la segunda se demuestra de forma análoga. Escribimos $f(z) = u(x, y) + i v(x, y)$. Entonces, por definición de derivada de Wirtinger y la regla de la cadena para derivadas parciales, se tiene que

$$2 \frac{\partial}{\partial z}(g \circ f)(z) = \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) (g \circ f)(z) = \frac{\partial g}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial g}{\partial v} \frac{\partial v}{\partial x} - i \frac{\partial g}{\partial u} \frac{\partial u}{\partial y} - i \frac{\partial g}{\partial v} \frac{\partial v}{\partial y} = (*).$$

Aplicando ahora el `lem-derivadas-parciales-wirtinger`/Lema 1, tenemos que

$$\begin{aligned} (*) &= \left(\frac{\partial g}{\partial z} + \frac{\partial g}{\partial \bar{z}} \right) \frac{\partial u}{\partial x} + i \left(\frac{\partial g}{\partial z} - \frac{\partial g}{\partial \bar{z}} \right) \frac{\partial v}{\partial x} - i \left(\frac{\partial g}{\partial z} + \frac{\partial g}{\partial \bar{z}} \right) \frac{\partial u}{\partial y} + \left(\frac{\partial g}{\partial z} - \frac{\partial g}{\partial \bar{z}} \right) \frac{\partial v}{\partial y} \\ &= \frac{\partial g}{\partial z} \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} - i \frac{\partial u}{\partial y} + \frac{\partial v}{\partial y} \right) + \frac{\partial g}{\partial \bar{z}} \left(\frac{\partial u}{\partial x} - i \frac{\partial v}{\partial x} - i \frac{\partial u}{\partial y} - \frac{\partial v}{\partial y} \right) \\ &= \frac{\partial g}{\partial z} \left(2 \frac{\partial u}{\partial z} + 2i \frac{\partial v}{\partial z} \right) + \frac{\partial g}{\partial \bar{z}} \left(2 \frac{\partial u}{\partial \bar{z}} + 2i \frac{\partial v}{\partial \bar{z}} \right) = 2 \frac{\partial g}{\partial z}(f(z)) \frac{\partial f}{\partial z}(z) + 2 \frac{\partial g}{\partial \bar{z}}(f(z)) \frac{\partial \bar{f}}{\partial \bar{z}}(z). \end{aligned}$$

Luego el resultado se sigue dividiendo entre 2. ■